Course Name:	***Digital Logic Design***	
Course Code:	CS-1104	
Disciplines:	BS (Cyber Security)	
Effective:	Batch 2023 and onwards	
Assessment:	10% Sessional, 10% Mid Semester Exam, 30% Final Semester Exam	
Marks:	**Theory:** 50	**Practical:** 50
Credit Hours:	2	1
Teaching Scheme:	2 Hr/Week	3 Hr/Week
Pre-requisites:	None	

Course Introduction:

The course introduces the concept of digital logic, gates and the digital circuits. Further, it focuses on the design and analysis combinational and sequential circuits. It also serves to familiarize the student with the logic design of basic computer hardware components.

Course Outline:

Number Systems, Logic Gates, Boolean Algebra, Combination logic circuits and designs, Simplification Methods (K-Map, Quinn Mc-Cluskey method), Flip Flops and Latches, Asynchronous and Synchronous circuits, Counters, Shift Registers, Counters, Triggered devices & its types. Mealy machines and Moore machines. Binary Arithmetic and Arithmetic Circuits, Memory Elements, State Machines. Introduction Programmable Logic Devices (CPLD, FPGA) Lab Assignments using tools such as Verilog HDL/VHDL, MultiSim.

CLO No.	**Course Learning Outcomes**	**Bloom Taxonomy**
CLO-1	Acquire knowledge related to the concepts, tools and techniques for the design of digital electronic circuits	C2 (Explain)
CLO-2	Demonstrate the skills to design and analyze both combinational and sequential circuits using a variety of techniques	C3 (Demonstrate)
CLO-3	Apply the acquired knowledge to simulate and implement small-scale digital circuits	C3 (Apply)
CLO-4	Understand the relationship between abstract logic characterizations and practical electrical implementations.	C4 (Use)

Reference Materials:

1. Digital Fundamentals by Floyd, 11/e.
2. Fundamental of Digital Logic with Verilog Design, Stephen Brown, 2/e